

Machine Learning with PySpark

Introduction

In this assignment, you will gain hands-on experience with machine learning using **PySpark MLlib**, Apache Spark's scalable machine learning library. You will learn how to load datasets into Spark DataFrames, preprocess data, build machine learning pipelines, train models, and evaluate their performance. The assignment consists of one **supervised learning** task and one **unsupervised learning** task using two popular real-world datasets.

Required PySpark Libraries

You are expected to use the appropriate PySpark libraries where applicable. The following modules will be useful for completing the assignment:

- `SparkSession` for creating and managing Spark applications.
- `VectorAssembler` for combining multiple feature columns into a single feature vector.
- `StringIndexer` for converting categorical variables into numeric indices.
- `OneHotEncoder` (optional) for encoding categorical features.
- `StandardScaler` (optional) for feature scaling before clustering.
- `LogisticRegression` or `RandomForestClassifier` for supervised learning.
- `KMeans` for unsupervised learning.
- `Pipeline` (optional) for organizing preprocessing and model training.
- `MulticlassClassificationEvaluator` or `BinaryClassificationEvaluator` for evaluating classification models.

You may use additional PySpark MLlib classes if necessary and take AI help as long as you can explain the code and process.

Part A: Supervised Learning – Titanic Survival Prediction

Dataset

Titanic – Machine Learning from Disaster (Kaggle)

The Titanic dataset contains information about passengers aboard the RMS Titanic, including demographic information, ticket details, travel class, and whether each passenger survived the disaster. Your task is to build a machine learning model that predicts passenger survival based on the available features.

Objective

Develop a supervised machine learning model using PySpark to predict whether a passenger survived the Titanic disaster.

Expected Tasks

1. Download the Titanic dataset from Kaggle and load the CSV file into a Spark DataFrame.
2. Explore the dataset by displaying its schema and basic statistics.
3. Identify and appropriately handle missing values.
4. Select suitable input features for prediction.
5. Convert categorical attributes (such as Sex and Embarked) into numerical representations using appropriate encoding techniques.
6. Combine all selected features into a single feature vector using `VectorAssembler`.
7. Split the dataset into training and testing sets.
8. Train **either** a Logistic Regression classifier **or** a Random Forest classifier.
9. Generate predictions on the test dataset.
10. Evaluate the model using suitable evaluation metrics such as Accuracy, Precision, Recall, or F1-score.
11. Display sample prediction results and briefly discuss the model's performance.

Deliverables

- Well-documented PySpark source code.
- Console output showing the model evaluation results.
- A short discussion (approximately one paragraph) describing the obtained accuracy and any preprocessing decisions made.

Part B: Unsupervised Learning – Mall Customer Segmentation

Dataset

Mall Customer Segmentation Data (Kaggle)

This dataset contains demographic and spending information of shopping mall customers. Unlike the previous task, there are no class labels. Instead, your objective is to discover natural customer groups using clustering techniques.

Objective

Use the K-Means clustering algorithm to group customers with similar characteristics based on their age, annual income, and spending score.

Expected Tasks

1. Download the Mall Customer Segmentation dataset from Kaggle and load it into a Spark DataFrame.
2. Explore the dataset by displaying the schema and descriptive statistics.
3. Select appropriate numerical features for clustering (e.g., Age, Annual Income, and Spending Score).
4. Create a feature vector using `VectorAssembler`.
5. Optionally normalize or standardize the selected features using `StandardScaler`.
6. Apply the K-Means clustering algorithm with either $k = 3$ or $k = 5$.
7. Display the predicted cluster assignment for each customer.
8. Print the cluster centers obtained from the trained model.
9. Briefly interpret the characteristics of the generated customer clusters.

Deliverables

- Well-documented PySpark source code.
- Output showing cluster assignments.
- Cluster center values generated by the K-Means model.
- A short discussion (approximately one paragraph) describing the discovered customer groups and the effect of the chosen value of k .

Submission Requirements

Submit a single compressed folder containing:

- PySpark source code for Part A.
- PySpark source code for Part B.
- A PDF report containing:
 - Name and ID.
 - Screenshots of important outputs.
 - Model evaluation results.
 - Cluster centers.
 - Brief discussions for both assignments.

Your code should be properly commented and organized so that another person can easily understand the implementation.